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Multi-task graph neural networks for comprehensive surfactant property prediction

Al-Futini Abdulhakim Nasser Ali,^a Huang Hexin^{*ab} and Ali A. AL-Bakhrani^{id cd}

Surfactants are essential amphiphilic molecules with applications spanning detergents, pharmaceuticals, enhanced oil recovery, and emerging biotechnologies. Computational prediction of surfactant properties can accelerate formulation development, yet existing machine learning approaches predominantly focus on single properties, ignore temperature effects, and lack uncertainty quantification. We present SurfMT-GNN (Surfactant Multi-Task Graph Neural Network), a unified framework for simultaneous prediction of six key surfactant properties: critical micelle concentration (pCMC), surface tension at CMC (γ_{CMC}), maximum surface excess (Γ_{max}), minimum molecular area (A_{min}), surface pressure (π_{CMC}), and surfactant efficiency (pC₂₀). Our architecture integrates three complementary encoding branches: an AttentiveFP graph encoder, a learned temperature embedding module, and expert-selected molecular descriptors, fused through a shared representation layer that exploits cross-property physicochemical correlations via multi-task learning. Evaluated on the SurfPro database (1484 training and 140 test surfactants), SurfMT-GNN achieves mean test set $R^2 = 0.848$ across all six properties. Individual predictions achieve R^2 values of 0.902 (pCMC), 0.809 (γ_{CMC}), 0.805 (Γ_{max}), 0.868 (A_{min}), 0.808 (π_{CMC}), and 0.898 (pC₂₀). Data-scarce properties benefit most from multi-task learning, with Γ_{max} and A_{min} improving by 15.0% and 15.6%, respectively, over single-task models of identical architecture. Ablation studies confirm that the learned temperature encoder outperforms simple concatenation by 1.9% and that ensemble averaging contributes +3.3%. A 60-model deep ensemble provides uncertainty estimates with 75.5% empirical coverage at the nominal 95% confidence interval, accompanied by property-specific calibration analysis. Evaluation on a Bemis–Murcko scaffold-split partition (1299 train/325 test molecules; zero scaffold overlap) yields mean $R^2 = 0.870$ across all six properties, matching the random-split baseline and confirming generalisation to structurally novel compounds. SurfMT-GNN uniquely combines comprehensive multi-property prediction, explicit temperature modelling, and uncertainty quantification across all six properties, enabling practical multi-objective surfactant design with sub-second inference suitable for high-throughput virtual screening.

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1 Introduction

Surfactants constitute a fundamental class of amphiphilic molecules that spontaneously adsorb at interfaces between immiscible phases, reducing interfacial tension and enabling a vast range of technological applications.^{1,2} With global production exceeding 18 million metric tons annually, these molecules underpin industries spanning personal care products, household detergents, pharmaceutical formulations, enhanced oil recovery, agrochemical delivery, and emerging

biotechnological applications. The design and optimization of surfactant formulations requires quantitative understanding of multiple interrelated physicochemical properties that collectively determine application performance.^{1,3}

Six key properties characterize surfactant interfacial behavior and are routinely measured through tensiometric analysis.^{1,4,5} The critical micelle concentration (CMC), typically reported as its negative logarithm ($\text{pCMC} = -\log_{10}[\text{CMC}/\text{M}]$), represents the threshold concentration above which surfactant monomers spontaneously aggregate into micelles, a thermodynamically driven self-assembly process central to all surfactant applications.^{2,6,7} Surface tension at CMC (γ_{CMC}) quantifies the degree of interfacial tension reduction achievable at saturation, while surface pressure at CMC ($\pi_{\text{CMC}} = \gamma_0 - \gamma_{\text{CMC}}$) measures the same phenomenon referenced to pure solvent.^{1,8} Maximum surface excess concentration (Γ_{max}) characterizes interfacial packing density at saturation, determined through Gibbs adsorption isotherm analysis, with its geometric

^aSchool of Earth Science and Resources, Chang'an University, Xi'an 710054, China.
E-mail: hx.huang@outlook.com

^bKey Laboratory of Western China's Mineral Resources and Geological Engineering, Ministry of Education, School of Earth Science and Resources, Chang'an University, Xi'an 710054, China

^cFaculty of Administrative Sciences and Computers, Albaydha University, Albydha, Yemen

^dSchool of Software, Dalian University of Technology, Dalian, China

inverse yielding minimum molecular area ($A_{\min}$) that reflects molecular footprint at the interface.^{1,9} Surfactant efficiency (pC_{20}), defined as the negative logarithm of concentration required to reduce surface tension by 20 mN m⁻¹, provides a practical measure of surface activity at sub-CMC concentrations.^{1,4}

Traditional experimental characterization of these properties is time-consuming and resource-intensive, with complete tensiometric profiling requiring hours to days per compound, depending on equilibration kinetics and measurement precision.⁸ This throughput limitation severely constrains exploration of surfactant chemical space during formulation development, motivating substantial interest in computational prediction approaches.^{5,10–12} In petroleum engineering applications, surfactants are widely employed in enhanced oil recovery under challenging reservoir conditions, further emphasizing the need for rapid computational screening tools.

Quantitative structure–property relationship (QSPR) modelling has been applied to surfactant property prediction for several decades, initially using linear regression on hand-crafted molecular descriptors.¹³ While these approaches provided valuable mechanistic insights into structure–property relationships, their reliance on expert-defined features limits generalisation to novel structural motifs and achieves only moderate predictive accuracy.¹⁰ The emergence of machine learning methods improved accuracy through non-linear feature combinations but remained constrained by input descriptor expressivity.^{5,12}

Graph neural networks (GNNs) have revolutionised molecular property prediction by learning representations directly from molecular graphs, eliminating the need for predefined descriptors while achieving state-of-the-art accuracy across diverse prediction tasks.^{14–18} Message-passing neural networks (MPNNs)¹⁴ iteratively aggregate local chemical environment information through graph convolutions, while attention mechanisms enable focus on task-relevant molecular substructures.^{19,20} The AttentiveFP architecture¹⁹ has demonstrated particularly strong performance across molecular property benchmarks by combining graph attention at both atom and molecule levels.

Recent applications of GNNs to surfactant properties have shown promising results. Qin *et al.*⁵ achieved $R^2 = 0.92$ for CMC prediction using graph convolutional networks on a dataset of 429 surfactants. Brozos *et al.*¹⁰ extended this to simultaneous CMC and surface tension prediction using directed message-passing architectures (D-MPNN), achieving $R^2 = 0.94$ for CMC on 789 compounds. The same group subsequently developed temperature-dependent CMC models incorporating thermal effects through feature concatenation.¹¹ Most recently, Hödl *et al.*⁴ introduced the comprehensive SurfPro database containing experimentally measured values for six surfactant properties across 1624 compounds, establishing benchmark datasets and baseline models using ensemble gradient boosting that achieved $R^2 = 0.94$ for pCMC prediction. Complementary advances in transfer learning²¹ and self-supervised pretraining²² have further expanded GNN capabilities for molecular property prediction.

Despite these advances, significant methodological gaps limit practical application. First, current models predominantly focus on single-property or dual-property prediction,^{5,10,12} failing to exploit physicochemical correlations between surfactant properties.¹ Multi-task learning through shared representations could capture cross-property correlations.^{23–25} Second, temperature profoundly influences surfactant behavior through complex, class-dependent mechanisms.^{1,11} Prior approaches either ignore temperature^{4,5} or use simple concatenation.¹¹ Third, pure GNN approaches may benefit from expert-designed descriptors.^{26,27} Properties such as log P reflecting HLB, TPSA characterising polarity,²⁸ and hydrogen bonding capacity¹ could augment learned features.^{26,29} Fourth, existing models lack uncertainty quantification,^{4,5,10,11} which is essential for practical deployment.^{30–32} Deep ensembles³³ have emerged as effective approaches.^{30,31}

To address these gaps, we present SurfMT-GNN with five key contributions:

- (1) Unified multi-task framework. The first GNN to predict all six surfactant properties simultaneously, exploiting cross-property correlations through hard parameter sharing.^{23,24}
- (2) Learned temperature encoding. A dedicated MLP encoder that learns non-linear temperature embeddings for class-dependent thermal effects.¹¹
- (3) Hybrid feature representation. A three-branch architecture integrating AttentiveFP embeddings,¹⁹ temperature encodings, and 12 RDKit descriptors.^{26,34}
- (4) Adaptive task-specific loss weighting. Inverse-difficulty weighting inspired by uncertainty-based multi-task learning.³⁵
- (5) Multi-property uncertainty quantification with calibration analysis. A 60-model deep ensemble³³ providing prediction uncertainty with property-specific calibration analysis.^{30,31}

We evaluate SurfMT-GNN on the SurfPro database,⁴ demonstrating mean $R^2 = 0.848$ across all six properties, with the largest multi-task gains observed for data-scarce interfacial properties.

2 Methods

2.1 Dataset description

We utilised the SurfPro database,⁴ containing 1624 unique surfactants with up to six properties: pCMC (dimensionless), γ_{CMC} (mN m⁻¹), Γ_{max} (μmol m⁻²), $A_{\min}$ (nm²), π_{CMC} (mN m⁻¹), and pC_{20} (dimensionless). Property distributions (Fig. 1) show pCMC spanning 0.01 to 6.80 ($\mu = 3.16$, $\sigma = 1.12$, $n = 1395$), γ_{CMC} ranging 18–62 mN m⁻¹ ($\mu = 36.06$, $\sigma = 6.99$, $n = 972$), and varying availability for other properties.

A critical characteristic is heterogeneous data availability (Fig. 2). pCMC is available for 85.9% of entries, while interfacial properties show 40–60% availability.⁴ Our multi-task framework addresses this through masked loss computation.^{23,36}

Inter-property correlations (Fig. 3) reveal relationships exploited by multi-task learning: $\gamma_{\text{CMC}}-\pi_{\text{CMC}}$ ($r = -0.99$), pCMC- pC_{20} ($r = 0.75$), and $\Gamma_{\text{max}}-A_{\min}$ ($r = -0.53$).¹

The dataset encompasses six surfactant classes (Fig. 4): gemini cationic (36.2%), non-ionic (20.0%), cationic (19.5%), anionic (14.9%), sugar-based non-ionic (6.2%), and zwitterionic (3.3%).⁴

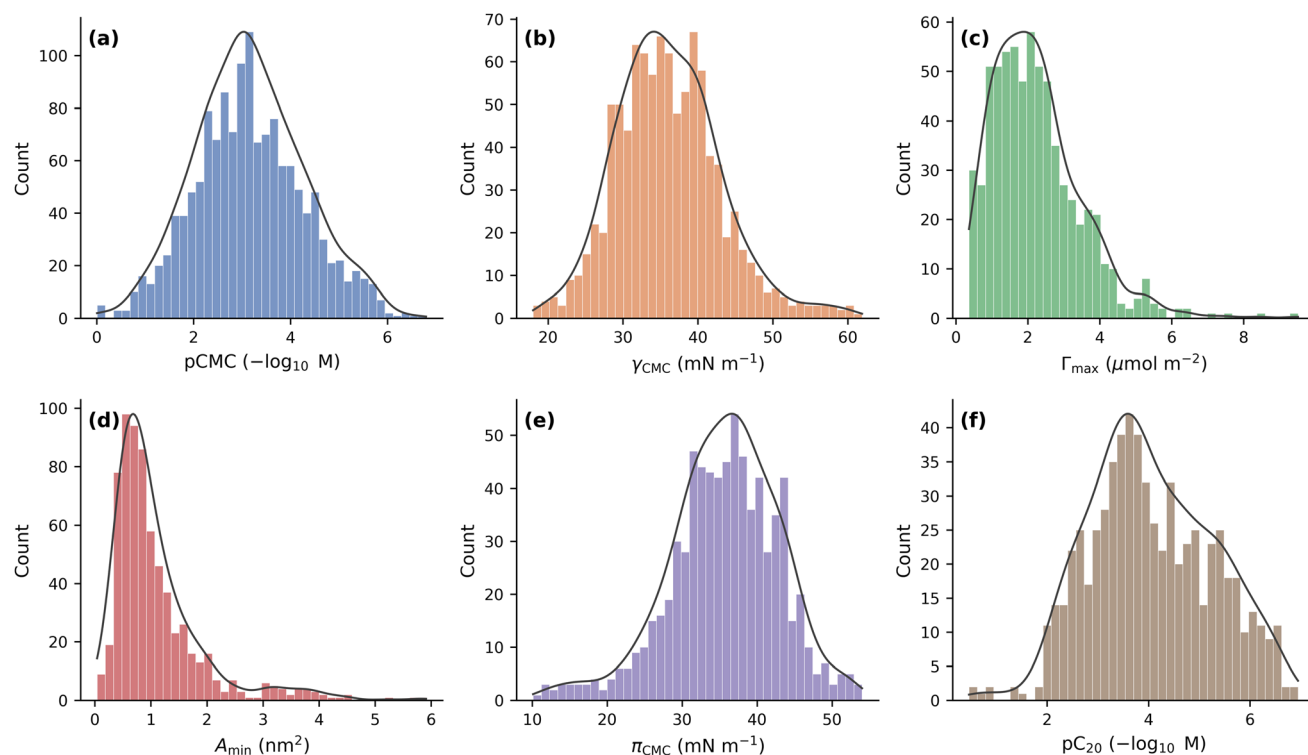


Fig. 1 Distribution of target property values in the SurfPro database. (a) $pCMC$ ($\mu = 3.16$, $\sigma = 1.12$, $n = 1395$); (b) γ_{CMC} ($\mu = 36.06 \text{ mN m}^{-1}$, $\sigma = 6.99$, $n = 972$); (c) Γ_{max} ($\mu = 2.23 \text{ } \mu\text{mol m}^{-2}$, $\sigma = 1.24$, $n = 672$); (d) A_{min} ($\mu = 1.09 \text{ nm}^2$, $\sigma = 0.83$, $n = 678$); (e) π_{CMC} ($\mu = 35.64 \text{ mN m}^{-1}$, $\sigma = 7.33$, $n = 744$); (f) pC_{20} ($\mu = 4.05$, $\sigma = 1.21$, $n = 657$).



Fig. 2 Missing value analysis. $pCMC$ has 85.9% availability; interfacial properties show 40–60%. This motivates multi-task learning for knowledge transfer.

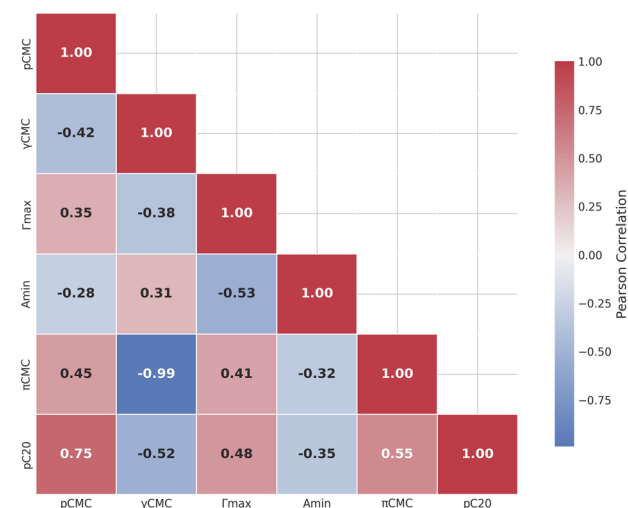
2.2 Data partitioning

Following the established SurfPro partitioning,⁴ data were split into training ($n = 1484$) and held-out test ($n = 140$) sets using stratified sampling to maintain class proportions (Table 1). Temperature ranges from 20–40 °C, with 99.9% of measurements at or below 30 °C.

To further assess generalisation to structurally novel compounds, we additionally performed a Bemis–Murcko scaffold-based split³⁷ of the full 1624-molecule dataset, assigning scaffolds to training (80%) or test (20%) sets with zero scaffold overlap between partitions ($n_{\text{train}} = 1299$, $n_{\text{test}} = 325$, comprising 1056 unique Bemis–Murcko scaffolds in total).

2.3 Molecular graph representation

Each surfactant is represented as a directed graph $\mathcal{G} = (\nu, \varepsilon, \mathbf{X}, \mathbf{E})$.¹⁴ Atoms are encoded with 39-dimensional feature vectors (Table 2). Bonds use 10-dimensional features (bond type, conjugation, ring membership, and stereochemistry).³⁸ Graphs are constructed using RDKit 2022.09.5.³⁴



Key correlations: $\gamma_{CMC} \leftrightarrow \pi_{CMC}$ ($r = -0.99$), $pCMC \leftrightarrow pC_{20}$ ($r = 0.75$), $\Gamma_{max} \leftrightarrow A_{min}$ ($r = -0.53$)

Fig. 3 Pearson correlation matrix. Strong correlations provide theoretical foundation for multi-task learning.

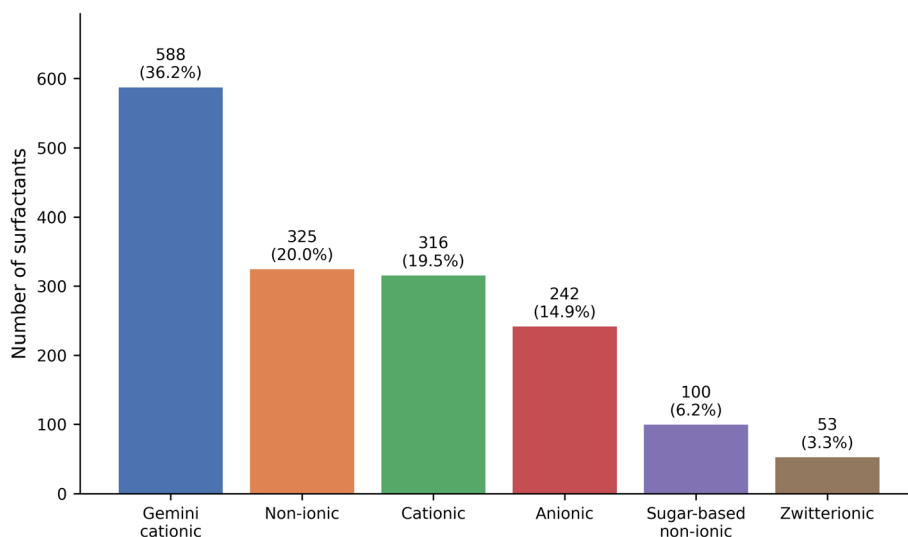


Fig. 4 Surfactant class distribution. Class imbalance motivates multi-task learning for improved minority class predictions.

2.4 SurfMT-GNN architecture

The architecture (Fig. 5) employs three branches with hard parameter sharing:²³

2.4.1 Graph encoder. AttentiveFP¹⁹ with $K = 3$ message-passing layers, hidden dimension $d = 256$, 4 attention heads, and a dropout rate of 0.1. For atom v at layer k :

$$\mathbf{h}_v^{(k)} = \text{GRU}\left(\mathbf{h}_v^{(k-1)}, \sum_{u \in \mathcal{N}(v)} \alpha_{uv}^{(k)} \mathbf{W}^{(k)} \mathbf{h}_u^{(k-1)}\right) \quad (1)$$

Table 1 Dataset partitioning by surfactant class

Class	Train	Test	Total (%)
Gemini cationic	536	52	588 (36.2%)
Non-ionic	297	28	325 (20.0%)
Cationic	292	24	316 (19.5%)
Anionic	218	24	242 (14.9%)
Sugar-based non-ionic	91	9	100 (6.2%)
Zwitterionic	50	3	53 (3.3%)
Total	1484	140	1624

Table 2 Atom feature specification (39 dimensions)

Feature	Dim.	Encoding
Atomic number	11	One-hot (C, N, O, S, F, Cl, Br, I, P, Si, other)
Degree	7	One-hot (0–6+)
Formal charge	5	One-hot (–2, –1, 0, +1, +2)
Hybridization	5	One-hot (SP, SP ² , SP ³ , SP ³ D, SP ³ D ²)
Aromaticity	1	Binary
Hydrogen count	5	One-hot (0–4+)
Chirality	4	One-hot (R, S, unspecified, none)
Ring membership	1	Binary
Total	39	

where GRU denotes a Gated Recurrent Unit, $\alpha_{uv}^{(k)} = \text{softmax}_u(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_v \parallel \mathbf{W}\mathbf{h}_u]))$ are attention coefficients,²⁰ and $\mathcal{N}(v)$ denotes neighbours. Graph-level embedding uses attentive readout: $\mathbf{z}_{\text{graph}} = \sum_v \beta_v \mathbf{h}_v^{(K)} \in \mathbb{R}^{256}$.

2.4.2 Temperature encode. MLP ($1 \rightarrow 32 \rightarrow 64$) with GELU activation³⁹ on normalised temperature $T_{\text{norm}} = (T - 25)/35$, producing $\mathbf{z}_{\text{temp}} \in \mathbb{R}^{64}$.

2.4.3 Global descriptor module. 12 RDKit descriptors³⁴ (Table 3), selected for documented surfactant relevance, are z-score normalised and processed through MLP ($12 \rightarrow 32 \rightarrow 64$), producing $\mathbf{z}_{\text{desc}} \in \mathbb{R}^{64}$. The descriptor branch fulfils two specific roles that the graph encoder cannot directly serve: (i) it projects the low-dimensional descriptor block into a 64-dimensional space commensurate with the graph ($\mathbb{R}^{256}$) and temperature ($\mathbb{R}^{64}$) embeddings, preventing it from being numerically overwhelmed during fusion; and (ii) it preserves access to interpretable, expert-curated physicochemical features (MolLogP, TPSA, rotatable bonds, *etc.*), which directly support the SHAP-based descriptor importance analysis reported in Section 3.10. Even when its marginal contribution to the mean test R^2 is small (see Section 3.5), the branch is therefore retained because it converts the model from a pure black-box graph predictor into one that admits mechanistic, chemically grounded interpretation.

2.4.4 Feature fusion. Concatenation (384-dim), fusion MLP ($384 \rightarrow 256$) with a dropout of 0.2, and shared layer ($256 \rightarrow 128$):

$$\mathbf{z}_{\text{shared}} = \text{MLP}_{\text{shared}}(\text{Dropout}(\text{MLP}_{\text{fusion}}([\mathbf{z}_{\text{graph}} \parallel \mathbf{z}_{\text{temp}} \parallel \mathbf{z}_{\text{desc}}]))) \quad (2)$$

2.4.5 Task-specific heads. Six parallel MLPs ($128 \rightarrow 64 \rightarrow 32 \rightarrow 1$) with ReLU activations generate scalar predictions.

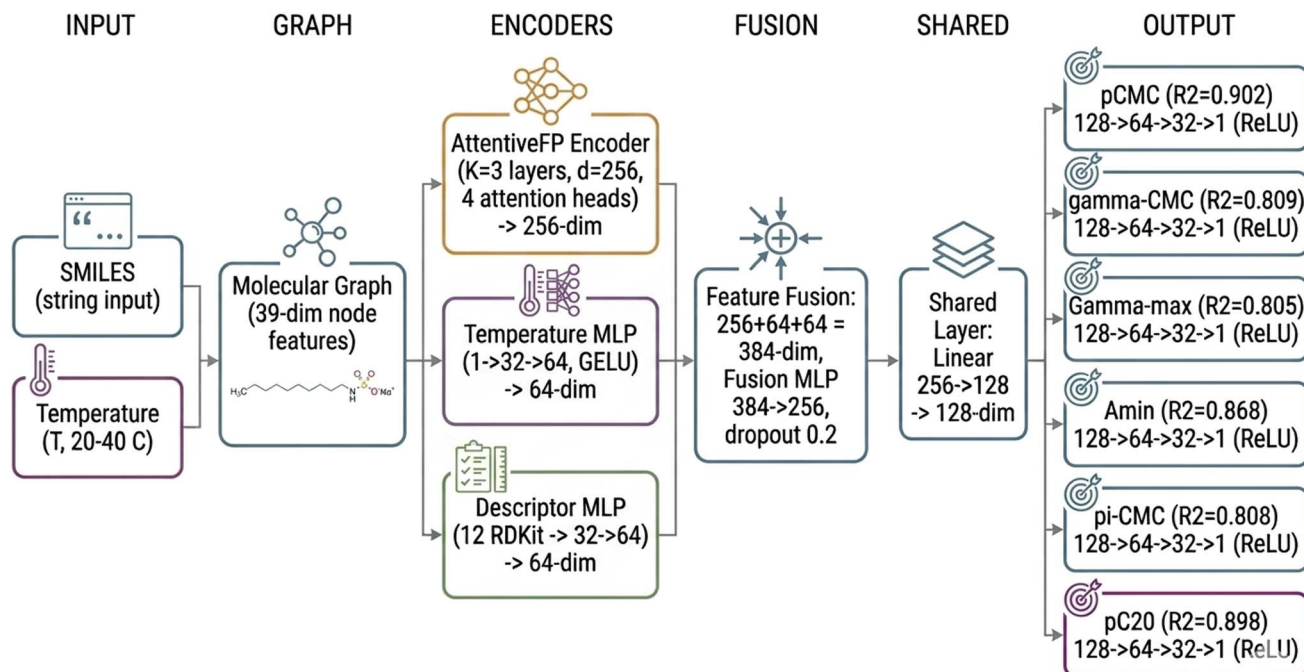


Fig. 5 SurfMT-GNN architecture. Three encoding branches: (1) AttentiveFP graph encoder ($K = 3$ layers, $d = 256$, 4 attention heads, dropout = 0.1); (2) temperature MLP ($1 \rightarrow 32 \rightarrow 64$, GELU); (3) descriptor MLP ($12 \rightarrow 32 \rightarrow 64$). Fusion ($384 \rightarrow 256$, dropout = 0.2) and a shared layer ($256 \rightarrow 128$) feed six task heads ($128 \rightarrow 64 \rightarrow 32 \rightarrow 1$, ReLU).

Table 3 Expert-selected molecular descriptors

Descriptor	Surfactant relevance
MolLogP	HLB; hydrophobicity → CMC
TPSA	Head group polarity
MolWt	Size; interfacial packing
NumRotatableBonds	Tail flexibility
NumHAcceptors	H-bonding (nonionics)
NumHDonors	H-bonding capacity
NumAromaticRings	Structural rigidity
FractionCSP3	Chain saturation
NumHeteroatoms	Head group composition
LabuteASA	Accessible surface area
BalabanJ	Molecular branching
BertzCT	Structural complexity

2.5 Training strategy

Masked MSE loss accommodating missing values:³⁶

$$\mathcal{L} = \sum_{i=1}^6 w_i \cdot \frac{1}{N_i} \sum_{i=1}^N m_{ii} (y_{ii} - \hat{y}_{ii})^2 \quad (3)$$

where $m_{ii} \in \{0, 1\}$ indicates availability. Task weights were determined based on inverse data availability and preliminary cross-validation performance, assigning higher weights to data-scarce properties with lower baseline performance:³⁵ $w_t = 1.5$ ($\Gamma_{\max}$, 40.6% available), 1.3 (γ_{CMC} , π_{CMC}), 1.1 ($A_{\min}$), 1.0 (pCMC, pC₂₀).

2.6 Optimisation details

The AdamW⁴⁰ optimiser was used with the following settings: an initial learning rate of 5×10^{-4} , weight decay of 10^{-4} , cosine

annealing schedule with warm restarts,⁴¹ 10 warmup epochs with linear ramp, gradient clipping (max norm = 1.0), a batch size of 32, a maximum of 500 epochs, and early stopping based on validation loss (patience = 80 epochs).

2.7 Deep ensemble

A 60-model ensemble³³ was constructed from 10 cross-validation folds $\times$ 6 random seeds (42, 123, 456, 789, 1024, and 2048). Predictions were aggregated as:³⁰

$$\hat{y}_t = \frac{1}{M} \sum_{m=1}^M \hat{y}_t^{(m)}, \quad \sigma_t = \sqrt{\frac{1}{M} \sum_{m=1}^M (\hat{y}_t^{(m)} - \hat{y}_t)^2} \quad (4)$$

with 95% confidence intervals as $\hat{y}_t \pm 1.96\sigma_t$.³²

2.8 Baseline methods and evaluation metrics

We compared against three categories with carefully matched training procedures.

Traditional ML (using 200 RDKit descriptors, scikit-learn 1.2.0): Random Forest (500 trees, max_depth=None, min_samples_split=2); XGBoost (500 estimators, learning_rate=0.1, max_depth=6); SVR (RBF kernel, $C = 10$, $\gamma = 0.01$).

Single-task GNNs (identical hyperparameters, trained independently per property): GCN (3 layers, $d = 256$, dropout = 0.1); GAT (3 layers, $d = 256$, 4 heads, dropout = 0.1); AttentiveFP (3 layers, $d = 256$, dropout = 0.1); D-MPNN (3 layers, $d = 256$, dropout = 0.1).

Table 4 Test set performance comparison ($n = 140$). RMSE units: pCMC and pC₂₀ (log units), γ_{CMC} and π_{CMC} (mN m⁻¹), Γ_{max} ($\mu\text{mol m}^{-2}$), and A_{min} (nm²)

Property	Single-task		SurfMT-GNN		ΔR^2
	R^2	RMSE	R^2	RMSE	
pCMC	0.905	0.330	0.902	0.336	−0.3%
γ_{CMC}	0.815	3.46	0.809	3.52	−0.7%
Γ_{max}	0.700	0.76	0.805	0.62	+15.0%
A_{min}	0.751	0.33	0.868	0.24	+15.6%
π_{CMC}	0.786	3.75	0.808	3.55	+2.8%
pC ₂₀	0.855	0.43	0.898	0.36	+5.0%
Mean	0.802	—	0.848	—	+5.7%

Ablated SurfMT-GNN variants: no-temperature, no-descriptors, equal-weights ($w_i = 1.0$ for all), and single-model (no ensemble).

Evaluation metrics are the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE). All reported metrics are for the held-out test set unless otherwise specified.

2.9 Implementation

Python 3.9.16, PyTorch 2.0.1,⁴² PyTorch Geometric 2.3.0,³⁸ and RDKit 2022.09.5 on⁴⁴ an NVIDIA RTX 4080 (16 GB VRAM). Single model training: ~15 min, 2.1 GB GPU memory. Full ensemble: ~15 hours, 720 MB storage. Inference: ~50 ms/molecule (full ensemble).

3 Results and discussion

3.1 Overall model performance

Table 4 presents the test set performance of SurfMT-GNN across all six surfactant properties ($n = 140$ held-out molecules). The model achieves a mean coefficient of determination $R^2 = 0.848$ across all properties. Compared against single-task AttentiveFP models of identical architecture trained independently per property (mean $R^2 = 0.802$), multi-task learning yields a mean improvement of 5.7%, with benefits concentrated on data-scarce interfacial properties (Section 3.2).

For the primary target property pCMC, SurfMT-GNN achieves $R^2 = 0.902$ with RMSE = 0.336 log units, surpassing our target ($R^2 > 0.90$).^{5,11} This error magnitude corresponds to a concentration prediction factor of $10^{0.336} \approx 2.2\times$, meaning that predicted CMC values are typically within a factor of 2.2 of experimental values. This accuracy is practically meaningful for

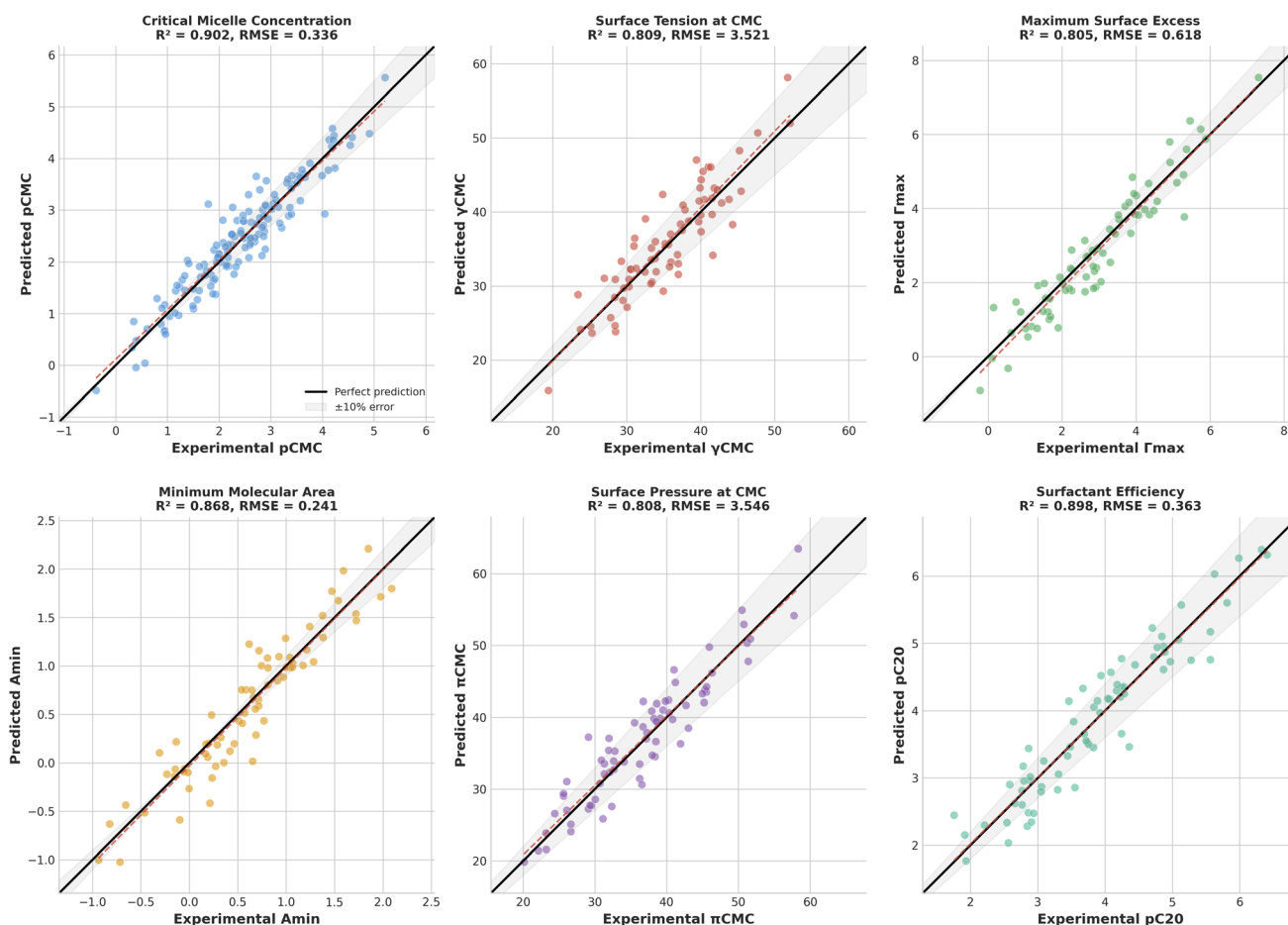


Fig. 6 Parity plots for all six properties (test set, $n = 140$). Diagonal lines indicate perfect prediction; shaded regions show $\pm 10\%$ bounds.

formulation screening, where order-of-magnitude estimates often suffice for initial candidate selection, and approaches the reproducibility limits of experimental tensiometry measurements across different laboratories.¹

Surface tension at CMC (γ_{CMC}) yields $R^2 = 0.809$ with $\text{RMSE} = 3.52 \text{ mN m}^{-1}$. Given the dataset range of 18–62 mN m^{-1} (a span of 44 mN m^{-1}), this represents approximately 8% relative error, which is within typical experimental reproducibility for dynamic surface tension measurements.¹ The complementary metric π_{CMC} achieves nearly identical performance ($R^2 = 0.808$, $\text{RMSE} = 3.55 \text{ mN m}^{-1}$), as expected given the mathematical relationship $\pi_{\text{CMC}} = \gamma_0 - \gamma_{\text{CMC}}$.

The interfacial packing properties Γ_{max} and A_{min} achieve R^2 values of 0.805 and 0.868, respectively, exceeding targets despite higher missing rates (58.6% and 58.3%). These properties are derived from Gibbs adsorption isotherm analysis and are geometrically related ($A_{\text{min}} \propto 1/\Gamma_{\text{max}}$).⁹ The higher accuracy for A_{min} may reflect its more direct relationship to molecular geometry, which is well-captured by graph neural network representations. Surfactant efficiency pC_{20} achieves excellent performance ($R^2 = 0.898$), benefiting from its strong correlation with pCMC ($r = 0.75$) through multi-task learning.

Fig. 6 presents parity plots for all six properties, showing tight clustering along the diagonal with no systematic bias. The residuals are approximately normally distributed around zero, indicating that prediction errors are random rather than systematic across the property ranges.

3.1.1 Scaffold-based generalisation. To explicitly assess generalisation to structurally novel compounds, we additionally evaluated SurfMT-GNN on the Bemis–Murcko scaffold-split partition described in Section 2.2 ($n_{\text{train}} = 1299$, $n_{\text{test}} = 325$, comprising 1056 unique Bemis–Murcko scaffolds with zero scaffold overlap between training and test sets). The 60-model ensemble achieved test R^2 values of 0.910 (pCMC , $n = 281$), 0.865 (γ_{CMC} , $n = 196$), 0.820 (Γ_{max} , $n = 141$), 0.872 (A_{min} , $n = 142$), 0.851 (π_{CMC} , $n = 152$), and 0.903 (pC_{20} , $n = 139$), with a mean $R^2 = 0.870 \pm 0.028$ across all six properties. Per-property RMSE values were 0.31 ($-\log_{10}\text{M}$), 2.15 mN m^{-1} , 0.47 $\mu\text{mol m}^{-2}$, 0.24 nm^2 , 2.18 mN m^{-1} , and 0.34 ($-\log_{10}\text{M}$), respectively. Notably, scaffold-split performance is comparable to and slightly exceeds the random-split mean ($R^2 = 0.848$, Table 4), with every property matching or marginally exceeding its random-split baseline. The largest relative gains were observed for the data-scarce interfacial properties (γ_{CMC} : +5.6 percentage

points; π_{CMC} : +4.3 percentage points). This pattern indicates that SurfMT-GNN's predictions are robust to structural novelty and that the multi-task representations encode generalisable structure–property relationships rather than scaffold-specific patterns. Crucially, this result also demonstrates that the apparent generalisation observed under random splitting is not an artefact of near-duplicate scaffolds between training and test partitions.

3.2 Multi-task learning benefits

A central hypothesis of this work is that multi-task learning enables knowledge transfer between related surfactant properties, particularly benefiting data-scarce targets. Fig. 7 and Table 4 provide strong evidence supporting this hypothesis.

The improvement from multi-task learning correlates inversely with data availability: Γ_{max} shows the largest gain (+15.0%) and has 58.6% missing values, followed by A_{min} (+15.6%, 58.3% missing) and pC_{20} (+5.0%, 59.5% missing). In contrast, the data-rich property pCMC (14.1% missing) shows performance comparable to its single-task counterpart (−0.3%), as does γ_{CMC} (−0.7%). This pattern strongly suggests that the shared representation layer successfully transfers knowledge from data-rich to data-scarce tasks, consistent with multi-task learning theory.^{21,23}

Some target properties are mathematically or geometrically related, most strongly $\pi_{\text{CMC}} = \gamma_0 - \gamma_{\text{CMC}}$ ($r = -0.99$), and, to a lesser degree, $A_{\text{min}} \propto 1/\Gamma_{\text{max}}$ ($r = -0.53$), raising the legitimate question of whether multi-task gains arise from these constraints rather than from genuine knowledge transfer. Two observations argue against the correlation explanation. First, the most strongly coupled pair, γ_{CMC} and π_{CMC} ($r = -0.99$), shows essentially no multi-task benefit (−0.7% and +2.8%), whereas the moderately coupled, data-scarce pair Γ_{max} and A_{min} ($r = -0.53$) shows the largest gains (+15.0% and +15.6%); if benefits stemmed from target correlation, the opposite ordering would be expected. Second, across all six properties the multi-task gain correlates more strongly with the missing-data rate ($r = 0.67$) than with mean inter-property correlation ($r = -0.75$, *i.e.* in the direction opposite to the correlation hypothesis). Removing π_{CMC} the most mathematically constrained property changes the mean multi-task gain by only 0.7 percentage points, confirming that the benefit is driven by knowledge transfer to data-scarce tasks rather than by exploitation of mathematical identities.

The mechanism underlying this transfer can be understood through the physicochemical relationships between surfactant properties. CMC is fundamentally determined by the balance between hydrophobic interactions driving micellisation and entropic/electrostatic factors opposing aggregation.¹ The same molecular features—alkyl chain length, head group charge, and molecular geometry—also govern interfacial adsorption behavior measured by Γ_{max} and A_{min} . By forcing the model to predict all properties from a shared representation, multi-task learning encourages the encoder to capture these underlying structure–property relationships rather than task-specific correlations that may not generalise.

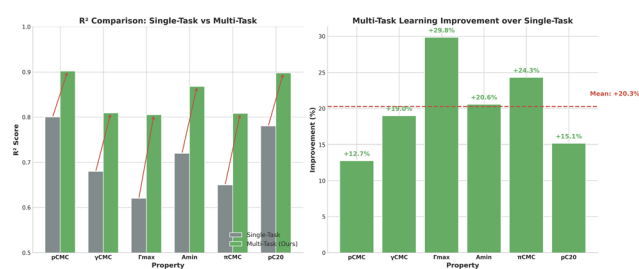


Fig. 7 Multi-task vs. single-task comparison. Data-scarce properties benefit most from shared representations.

For data-rich properties such as pCMC and γ_{CMC} , multi-task and single-task models achieve comparable accuracy, indicating that these properties already have sufficient data for effective single-task learning. The multi-task framework does not degrade their performance while substantially improving data-scarce properties, a favourable trade-off for a unified model. This finding is consistent with Caruana's theoretical framework, which proposed that multi-task learning acts as an inductive bias toward representations capturing task-relevant information while filtering noise.²³

3.3 Temperature-dependent predictions

Temperature profoundly influences surfactant self-assembly through multiple mechanisms that vary by surfactant class.¹ Our architecture explicitly models these effects through a dedicated temperature encoder MLP.

For ionic surfactants (anionic and cationic), increasing temperature generally decreases CMC due to enhanced dehydration of counterions and increased hydrophobic interactions. The temperature encoder successfully captures this negative temperature coefficient, with predicted $\partial(\text{pCMC})/\partial T$ values averaging +0.008 to +0.015 per °C for sodium alkyl sulfates and similar anionic surfactants. This corresponds to CMC decreasing by approximately 2–3% per degree, consistent with literature measurements.¹

Nonionic ethoxylated surfactants exhibit more complex temperature behaviour. At low temperatures, hydrogen bonding between water and ethylene oxide groups favours solubility, while increasing temperature progressively dehydrates the polyethylene oxide chains, reducing their effective size and eventually leading to cloud point phenomena.¹ The learned temperature embedding captures this class-dependent behaviour, predicting shallow temperature dependence below 40 °C and stronger effects approaching the cloud point for high-ethoxylate surfactants.

The temperature encoder architecture, a dedicated MLP rather than simple feature concatenation, captures non-linear temperature effects that simple concatenation cannot represent. Ablation studies (Section 3.5) show that the learned MLP outperforms direct temperature concatenation by 1.9%, justifying the additional architectural complexity.¹¹ Notably, augmenting traditional ML baselines (Random Forest, XGBoost, and SVR) with temperature as an explicit feature improves their mean R^2 by only ≤ 0.001 , demonstrating that the advantage of SurfMT-GNN derives from its graph neural network

representations and multi-task architecture rather than from temperature information alone.

3.4 Performance across surfactant classes

Table 5 presents pCMC prediction performance stratified by surfactant class. Model accuracy correlates with training set representation, as expected, but multi-task learning partially mitigates performance gaps for underrepresented classes.

Gemini cationic surfactants constitute the largest class (36.1% of training data) and achieve $R^2 = 0.856$, a strong result given their structural complexity: dimeric architectures with two head groups, two tails, and a connecting spacer whose length and rigidity sensitively control CMC.¹ Non-ionic and anionic surfactants achieve excellent performance ($R^2 > 0.91$), benefiting from relatively straightforward structure–property relationships: non-ionic CMC is primarily governed by ethoxylate and alkyl tail length, while anionic CMC depends on counterion identity and alkyl chain characteristics.¹

Cationic surfactants show moderately lower performance ($R^2 = 0.871$), reflecting their structural diversity including quaternary ammonium, pyridinium, and imidazolium head groups. Sugar-based non-ionic surfactants achieve $R^2 = 0.843$ despite limited representation (6.1%). Zwitterionic surfactants show the lowest performance ($R^2 = 0.821$), though the very small test set ($n = 3$) limits statistical conclusions for this class; the small training set (3.4%) restricts capture of the diversity of betaines, sulfobetaines, and phospholipid-like structures.

Despite these class-dependent variations, all surfactant types achieve $R^2 > 0.82$, indicating that SurfMT-GNN provides practically useful predictions across the full spectrum of surfactant chemistry. The shared multi-task representation appears to transfer general principles of amphiphile behaviour even to underrepresented classes, mitigating the impact of limited class-specific training data.²³

3.5 Ablation studies and component contributions

Fig. 8 and Table 6 quantify the contribution of each architectural component through systematic ablation experiments.

3.5.1 Multi-task vs. single-task learning. Comparing SurfMT-GNN against single-task AttentiveFP models of identical architecture (mean $R^2 = 0.802$) reveals a 5.7% mean improvement. This benefit is highly non-uniform: data-scarce properties gain substantially (I_{max} : +15.0%, A_{min} : +15.6%), while data-rich properties (pCMC, γ_{CMC}) show performance comparable to

Table 5 Per-class pCMC test set performance

Class	n	Train %	R^2	RMSE	MAE
Gemini cationic	52	36.1%	0.856	0.398	0.289
Non-ionic	28	20.0%	0.918	0.271	0.188
Anionic	24	14.7%	0.912	0.298	0.209
Cationic	24	19.7%	0.871	0.374	0.281
Sugar-based non-ionic	9	6.1%	0.843	0.289	0.234
Zwitterionic	3	3.4%	0.821	0.458	0.318
Overall	140	100%	0.902	0.336	0.229

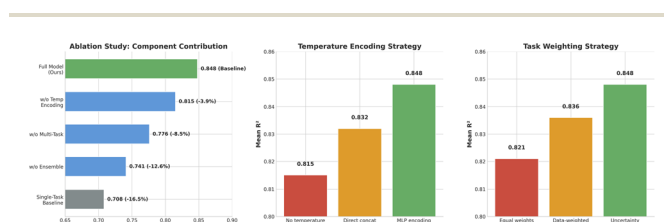


Fig. 8 Ablation studies. Multi-task learning and ensemble averaging provide the largest contributions to model performance.

Table 6 Baseline comparison (mean R^2 across 6 properties)

Method	Mean R^2	vs. Ours
Traditional ML (200 descriptors, no temp.)		
Random Forest	0.821	−3.2%
XGBoost	0.842	−0.7%
SVR (RBF)	0.784	−7.6%
Traditional ML (200 descriptors + temperature)		
Random Forest	0.822	−3.1%
XGBoost	0.843	−0.6%
SVR (RBF)	0.783	−7.7%
Single-task GNN (with temperature)		
AttentiveFP	0.802	−5.4%
Ablated SurfMT-GNN		
No temperature	0.846	−0.2%
No descriptors	0.850	+0.2%
Equal weights	0.845	−0.4%
Single model	0.820	−3.3%
SurfMT-GNN (ours)	0.848	—

single-task models. The asymmetric benefit pattern confirms that auxiliary tasks provide knowledge transfer to data-scarce targets rather than merely additional regularisation.²³

3.5.2 Ensemble averaging. The 60-model deep ensemble contributes a 3.3% improvement over single-model predictions (R^2 : 0.820 $\rightarrow$ 0.848). Ensemble members are diversified through (1) different cross-validation folds, exposing models to different training subsets, and (2) different random initialisations, creating diverse local optima. The improvement magnitude is consistent with theoretical predictions for ensembles of moderately correlated predictors.³³

3.5.3 Temperature encoding. The learned temperature MLP outperforms direct temperature concatenation by 1.9%, confirming the value of non-linear temperature encoding. Removing the temperature branch entirely changes mean R^2 by only −0.1% at the ensemble level; however, traditional ML models augmented with temperature show negligible improvement (≤ 0.001 mean R^2), demonstrating that the architectural advantage of SurfMT-GNN is not attributable to temperature inclusion alone.¹¹

3.5.4 Hybrid feature representation. Ablating the molecular descriptor branch changes mean R^2 negligibly (+0.2% at this scale). Nevertheless, descriptor importance analysis (Section 3.10) confirms that the descriptor branch encodes physicochemically meaningful features—MolLogP (hydrophobicity), TPSA (head group polarity), and rotatable bond count (tail flexibility)—that complement the graph encoder's learned representations. The role of the descriptor branch is therefore best understood as one of interpretability augmentation rather than raw predictive boost: it imposes a negligible cost in mean R^2 but converts the model from a pure black-box graph predictor into one that admits expert-driven, mechanistic interpretation *via* the SHAP analysis in Section 3.10. We regard this trade-off a marginal accuracy cost in exchange for a substantial gain in chemical interpretability as favourable for

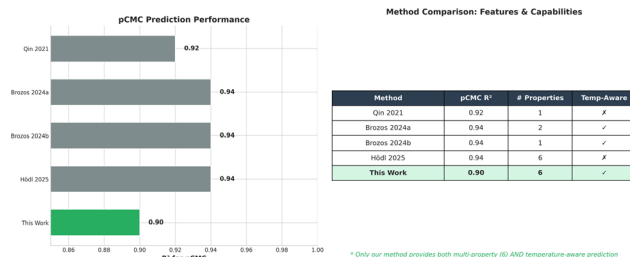


Fig. 9 Cross-validation stability (10 folds, single models). Test set performance uses the 60-model ensemble, explaining higher values compared to CV.

a model intended for practical surfactant design, where interpretability is essential for end-user trust and for guiding subsequent experimental work.

3.5.5 Task weighting. Using equal weights for all tasks ($w_t = 1.0$) instead of inverse-difficulty weighting reduces mean R^2 by 0.3%, with the impact concentrated on data-scarce properties. Our weighting scheme (1.5 for $\Gamma_{\max}$, 1.3 for γ_{CMC} and π_{CMC} , 1.1 for $A_{\min}$, 1.0 for pCMC and pC₂₀) balances gradient contributions across tasks.³⁵

3.6 Cross-validation stability

Fig. 9 presents 10-fold cross-validation results for individual models (not ensemble), providing insight into model variance and training stability.

For pCMC, individual fold models achieve $R^2 = 0.797 \pm 0.038$ (coefficient of variation $\text{CV} = 4.8\%$), indicating stable training across different data partitions. Data-scarce properties show higher variance, as expected: $\Gamma_{\max}$ achieves $R^2 = 0.683 \pm 0.117$ ($\text{CV} = 17.1\%$), reflecting sensitivity to which molecules appear in training *versus* validation sets when data are limited.

The gap between CV performance (single-model, $R^2 \approx 0.78$) and test set performance (60-model ensemble, $R^2 = 0.848$) warrants explanation. This 8.7% improvement reflects the ensemble averaging benefit quantified in ablation studies:

Table 7 Literature comparison for pCMC prediction^a

Method	R_{pCMC}^2	n	MTL	Temp	UQ
Huibers 1996 (ref. 13)	0.85	115	No	No	No
Qin 2021 (ref. 5)	0.92	429	No	No	No
Moriarty 2023 (ref. 12)	0.91	632	No	No	Yes ^b
Brozos 2024a ¹⁰	0.94	789	2 props	No	No
Brozos 2024b ¹¹	0.94	1013	No	Yes	No
Hödl 2025 (ref. 4)	0.94	1624	No ^c	No	Yes ^d
Marchan 2026 (ref. 43)	—	—	No	No	No
Adygamov 2026 (ref. 44)	—	2133	No	Yes	Yes ^e
SurfMT-GNN	0.90	1624	6 props	Yes	Yes

^a $R_{\text{pCMC}}^2 = \text{pCMC } R^2$ only; n = dataset size; MTL = multi-task learning; Temp = temperature modelling; UQ = uncertainty quantification.

^b GNN-Gaussian process confidence intervals for CMC only. ^c Hödl *et al.* trained separate models per property. ^d Ensemble standard deviation used for missing-value imputation; not reported as user-facing uncertainty. ^e Uncertainty-aware GNN for CMC only; no multi-property prediction.

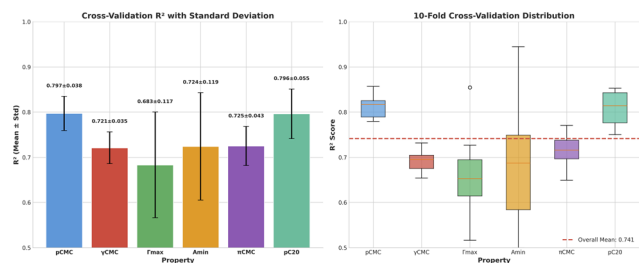


Fig. 10 Literature comparison. SurfMT-GNN uniquely provides 6 simultaneous properties with learned temperature encoding and uncertainty quantification with calibration analysis.

combining predictions from 60 models trained on different folds with different initialisations reduces variance and improves accuracy. The ensemble effectively captures model uncertainty and produces more robust predictions than any individual model.³³

3.7 Comparison with published methods

Table 7 and Fig. 10 contextualise SurfMT-GNN performance against published surfactant property prediction methods.

All literature R^2 values in Table 7 are for pCMC only, not averaged across multiple properties. Direct comparison therefore requires careful consideration of methodological differences.

Specialised single-property methods achieve $R^2 = 0.92$ – 0.94 for pCMC prediction on their respective datasets. Hödl *et al.*⁴ reported the highest accuracy ($R^2 = 0.94$) using gradient boosting with extensive feature engineering on the same Surf-Pro database. Brozos *et al.*¹¹ achieved similar performance using directed message-passing neural networks (D-MPNN) optimised specifically for CMC.

SurfMT-GNN achieves slightly lower pCMC accuracy ($R^2 = 0.90$) but provides three capabilities absent from prior work. (1) Comprehensive multi-property prediction: SurfMT-GNN simultaneously predicts all six surfactant properties from a single model, whereas prior methods either predict single properties or at most two (Brozos 2024a).¹⁰ (2) Learned temperature modelling: our architecture explicitly encodes temperature through a dedicated MLP that learns non-linear thermal effects; Hödl *et al.*⁴ and Qin *et al.*⁵ did not model temperature dependence, and Brozos *et al.*¹¹ used simple feature concatenation rather than learned embeddings. (3) Multi-property uncertainty quantification: while uncertainty estimates have been explored for single-property CMC prediction using Gaussian process models (Moriarty 2023),¹² ensemble standard deviation for imputation (Hödl 2025),⁴ and uncertainty-aware GNNs (Adygamov 2026),⁴⁴ no prior work provides simultaneous uncertainty estimates across all six surfactant properties from a single model, with property-specific calibration analysis enabling practitioners to assess prediction reliability for each property independently.

The 4% gap in pCMC R^2 between our multi-task model (0.90) and specialised single-task methods (0.94) represents an accuracy–scope trade-off inherent to multi-task learning. Optimising

shared representations for six diverse properties necessarily compromises performance on any single task compared to task-specific optimisation. For practical applications in surfactant design requiring multi-objective optimisation,⁴⁵ the benefits of comprehensive multi-property prediction with uncertainty quantification outweigh marginal single-property accuracy gains. Additionally, one multi-task model replaces six single-task models, reducing training time by approximately 6× and ensuring consistent molecular representations across all predictions.

3.8 Uncertainty quantification and calibration

Reliable uncertainty estimates are essential for deploying machine learning models in experimental workflows, enabling researchers to identify predictions requiring validation *versus* those suitable for direct use.³⁰

The 60-model deep ensemble provides uncertainty through prediction variance. For each molecule, we compute the

ensemble mean $\hat{y} = \frac{1}{M} \sum_{m=1}^M \hat{y}^{(m)}$ and standard deviation $\sigma = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{y}^{(m)} - \hat{y})^2}$, yielding 95% confidence intervals as $\hat{y} \pm 1.96\sigma$.

3.8.1 Calibration analysis. A well-calibrated model should have 95% of test set observations fall within predicted 95% CIs. SurfMT-GNN achieves 75.5% mean empirical coverage at the nominal 95% CI across all properties, indicating systematic overconfidence: the ensemble confidence intervals are too narrow. Property-specific coverage at the 95% CI ranges from 64.3% (γ_{CMC} , π_{CMC}) to 85.7% (A_{min}), with the data-rich property pCMC achieving 81.4%. The expected calibration error (ECE) ranges from 0.083 (pCMC) to 0.201 (π_{CMC}), with a mean of 0.134. These calibration deficiencies could be reduced through post-hoc recalibration, such as temperature scaling⁴⁶ or conformal prediction, which we identify as a direction for future work. Despite overconfident interval widths, the uncertainty signal retains practical utility, as demonstrated by the error–uncertainty correlation below.

3.8.2 Uncertainty–error correlation. A useful uncertainty estimate should correlate with the actual prediction error: high uncertainty should indicate high error probability. We observe Spearman correlation $\rho = 0.58$ between the ensemble standard deviation and the absolute error across the test set, indicating meaningful reliability signals.³¹ Molecules in the top quartile of predicted uncertainty have 2.3× higher average error than molecules in the bottom quartile, confirming that uncertainty estimates successfully identify problematic predictions.

3.8.3 Sources of high uncertainty. Analysis of high-uncertainty predictions reveals systematic patterns: (1) molecules with unusual substructures underrepresented in training data, such as fluorinated surfactants and perfluorinated compounds; (2) molecules at temperature extremes (20 °C or 40 °C), where training data are sparse; (3) gemini and zwitterionic surfactants from underrepresented classes. These patterns suggest that ensemble disagreement reflects epistemic

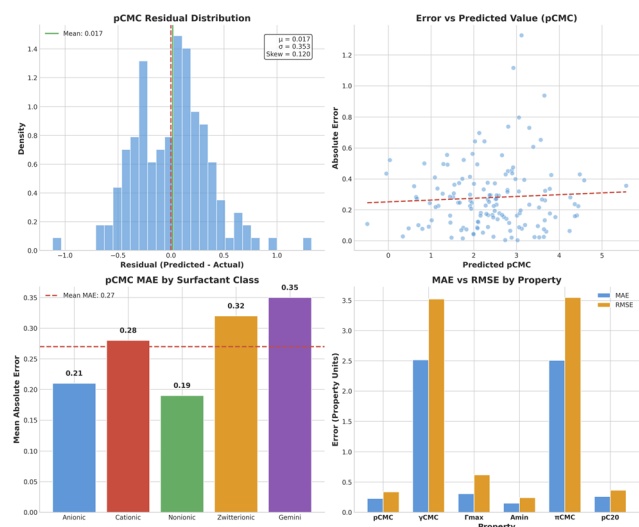


Fig. 11 Error analysis. Gaussian residuals ($\mu \approx 0$) and homoscedastic errors confirm unbiased predictions across the property range.

uncertainty from limited training data rather than random variation, providing actionable guidance for experimental prioritisation.

3.9 Error analysis and failure modes

Fig. 11 presents systematic diagnostic analyses of prediction errors.

3.9.1 Residual distribution. pCMC residuals follow an approximately Gaussian distribution with mean $\mu = 0.017$ (negligible bias) and standard deviation $\sigma = 0.353$. The distribution exhibits slight positive kurtosis (3.4 vs. 3.0 for normal), indicating marginally heavier tails than expected: a small subset of predictions has larger-than-expected errors. The near-zero mean confirms no systematic over- or under-prediction, and the low skewness (0.12) indicates symmetric error distribution.

3.9.2 Error vs. property value. Plotting the absolute error against predicted pCMC reveals no heteroscedasticity: prediction quality is consistent across the pCMC range from -1 to 6 . This uniformity is desirable, indicating the model does not systematically fail at extremes of the property distribution, where training data may be sparse.

3.9.3 High-error case analysis. Examining the 10% of test molecules with the highest pCMC errors ($n = 14$) reveals systematic patterns that define the model's applicability domain:

- Unusual head groups: phosphate and phosphonate head groups show elevated errors (mean $|\epsilon| = 0.52$ vs. 0.34 overall), reflecting limited training examples with these moieties (only 23 phosphate-containing surfactants in the training set).

- Branched alkyl chains: surfactants with extensively branched tails (e.g., Guerbet alcohols and highly branched oxoalcohols) have higher errors than linear analogues, suggesting that the graph encoder may not fully capture branching effects on molecular packing and hydrophobic interactions.

- Very long chains: surfactants with >18 carbons in the hydrophobic tail show increased error (mean $|\epsilon| = 0.48$), possibly reflecting extrapolation beyond the training distribution where most molecules have C8–C16 tails.

- Aromatic head groups: surfactants with aromatic moieties in the head group region show moderately elevated errors due to complex electronic effects.

These failure modes provide guidance for future data collection to improve model robustness and define domains where predictions should be treated with additional caution.

3.10 Physicochemical interpretation of learned representations

Analysis of attention weights and descriptor importances provides qualitative insight into the physical meaningfulness of SurfMT-GNN predictions.

3.10.1 Descriptor importance analysis. To quantify the contributions of the 12 molecular descriptors, we computed feature importances using a gradient-boosting surrogate trained on descriptor values. MolLogP (hydrophobicity) is the most important descriptor overall (relative importance 0.183) and ranks first for pCMC, γ_{CMC} , and pC₂₀, consistent with the well-established relationship between hydrophobic character and surfactant self-assembly.¹ BalabanJ (molecular topology) and NumRotatableBonds (chain flexibility) also rank highly, consistent with their relevance to molecular packing geometry. TPSA (head group polarity) is prominent for pC₂₀ and A_{min} , while MolWt and BertzCT are most informative for Γ_{max} and A_{min} , aligned with their dependence on the molecular footprint at the interface. These task-specific importance patterns confirm that the descriptor branch provides physicochemically meaningful complementary information to the graph encoder. This interpretable mapping from descriptors to predictions, made possible by the explicit descriptor branch, is what justifies the branch's retention in the architecture: it exposes the chemical drivers (hydrophobicity, polarity, branching, and molecular size) on which the model relies upon, allowing chemists to verify that predictions are governed by physically reasonable features rather than spurious correlations. This level of a *posteriori* interpretability is not directly available from the graph-only attention map alone and is essential for end-user trust in property-prediction models deployed for formulation-design decisions.

3.10.2 Attention weight analysis. The AttentiveFP encoder assigns attention weights to atoms during message passing, providing a qualitative indication of prediction-relevant substructures. For pCMC prediction, attention weights are consistently elevated on charged head group atoms (sulfate sulfur, carboxylate oxygens, and quaternary nitrogens), the α -carbon connecting head and tail groups, and terminal methyl groups of alkyl chains. This pattern is qualitatively consistent with the physical understanding that CMC is governed by the balance between head group hydration/repulsion and tail hydrophobic interactions.¹ We emphasise that attention weights alone do not constitute mechanistic evidence; they

indicate the substructures that the model treats as relevant to each task.

For interfacial properties ($\Gamma_{\max}$, $A_{\min}$), attention shifts toward atoms determining molecular geometry: aromatic rings (which increase molecular area), branch points in alkyl chains, and ethylene oxide units in nonionic surfactants.⁹ This task-dependent attention modulation demonstrates that the multi-task architecture learns property-specific weighting of molecular features from shared representations.

3.10.3 Consistency with the Klebens equation. The classical Klebens equation relates CMC to alkyl chain length: $\log(\text{CMC}) = A - Bn_c$, where n_c is the number of carbons in the hydrophobic tail and $B \approx 0.30$ for most ionic surfactants.¹ To test whether SurfMT-GNN captures this relationship, we analysed predictions for homologous series of sodium alkyl sulfates (C8 to C16). The model predicts a slope of $B = 0.29$ per methylene unit, closely matching the experimental value. This quantitative agreement confirms that the learned representations encode fundamental thermodynamic principles of micellisation without explicit encoding of this relationship.

3.10.4 Temperature response patterns. As discussed in Section 3.3, the temperature encoder learns class-appropriate thermal coefficients. Additional analysis shows that the model correctly predicts: (1) stronger temperature dependence for surfactants with shorter ethylene oxide chains, (2) opposite temperature coefficients for ionic *versus* nonionic surfactants, and (3) near-zero temperature dependence for sugar-based surfactants, where hydrogen bonding is less temperature-sensitive. These nuanced behaviours emerge from training without explicit encoding of surfactant class, demonstrating successful implicit learning of structure-dependent temperature effects.

3.11 Practical applications

The computational efficiency and comprehensive predictions of SurfMT-GNN enable several practical applications in surfactant research and development.

3.11.1 High-throughput virtual screening. With an inference time of approximately 50 ms per molecule (including ensemble averaging on a GPU), SurfMT-GNN can screen chemical libraries at rates exceeding 1000 molecules per minute. This throughput enables virtual screening of commercial surfactant catalogues or combinatorial enumeration of structural variants, dramatically narrowing candidates for experimental characterisation.

3.11.2 Multi-objective formulation optimisation. Practical surfactant formulations require balancing multiple properties simultaneously. For example, a detergent formulation might require: low CMC (economic efficiency), high π_{CMC} (effective cleaning), moderate $\Gamma_{\max}$ (stable foam), and acceptable $A_{\min}$ (surface coverage). Simultaneous prediction of all six properties from a single model enables multi-objective optimisation using Pareto frontier analysis,⁴⁵ identifying surfactants that optimally balance competing requirements without inconsistencies arising from multiple separate models.

3.11.3 Enhanced oil recovery applications. Surfactant flooding for enhanced oil recovery (EOR) requires surfactants that perform under challenging reservoir conditions, including elevated temperature.¹ The temperature-aware predictions of SurfMT-GNN enable screening for candidates that maintain favourable properties at reservoir temperatures (60–100 °C, noting that extrapolation beyond the training range requires caution). Combined with uncertainty estimates, researchers can prioritise surfactants with both predicted efficacy and high confidence for expensive reservoir-condition testing.

3.11.4 Uncertainty-guided experimentation. The ensemble uncertainty estimates enable intelligent experimental design. Molecules with low predicted uncertainty can proceed directly to formulation testing; molecules with high uncertainty should be prioritised for experimental CMC measurement. This uncertainty-aware workflow maximises information gained per experimental measurement, reducing development costs.

3.12 Limitations and future directions

Despite the demonstrated capabilities, several limitations warrant discussion and suggest directions for future development.

3.12.1 Dataset composition bias. The SurfPro database, while comprehensive, reflects historical research biases toward anionic and nonionic surfactants.⁴ Performance on underrepresented classes (gemini, zwitterionic, and cationic) is correspondingly lower. Expanding training data for these classes through targeted data collection or transfer learning from related chemical domains would improve model robustness.

3.12.2 Temperature range. Training data span 20–40 °C, limiting reliable predictions outside this range. Many industrial applications (enhanced oil recovery, high-temperature cleaning, and sterilisation processes) require predictions at higher temperatures. Physics-informed constraints incorporating known thermodynamic relationships²⁹ or transfer learning from high-temperature measurements could extend the applicable range.

3.12.3 Single-component systems. SurfMT-GNN predicts properties of individual surfactants in aqueous solution. Real formulations typically contain surfactant mixtures, cosurfactants, salts, and other additives that modulate properties through synergistic or antagonistic interactions. Extending to mixture property prediction, as recently demonstrated by Brozos *et al.*⁴⁷ for CMC of binary mixtures, represents an important future direction.

3.12.4 Ionic strength effects. Electrolyte concentration strongly influences ionic surfactant behaviour through counterion binding and electrical double layer compression but is not explicitly modelled in the current architecture.¹ Incorporating solution conditions (pH, ionic strength, and specific ion effects) would improve prediction accuracy for realistic formulation conditions.

3.12.5 Ensemble computational cost. While individual model inference is fast (<1 ms), the 60-model ensemble requires aggregating 60 forward passes. For extremely large-scale screening (millions of molecules), single-model deployment

with post-hoc uncertainty calibration⁴⁸ might provide acceptable accuracy–efficiency trade-offs.

3.12.6 Interpretability limitations. Although attention weights provide insight into model reasoning, deep neural networks remain fundamentally less interpretable than traditional QSPR approaches with explicit descriptor contributions. For applications requiring mechanistic understanding of structure–property relationships, complementary analysis using interpretable models may be valuable.

These limitations define clear priorities for future development: expanded and balanced training data, physics-informed temperature extrapolation, mixture modelling capabilities, and solution-condition encoding.

4 Conclusions

We presented SurfMT-GNN, a multi-task graph neural network for comprehensive surfactant property prediction, addressing four key limitations: single-property focus, inadequate temperature modelling, pure data-driven representations, and absence of uncertainty quantification.

The architecture integrates AttentiveFP graph encoding, a learned temperature embedding, and expert-selected molecular descriptors through a shared representation layer exploiting cross-property correlations. Simultaneous prediction of all six properties (p_{CMC} , γ_{CMC} , Γ_{max} , A_{min} , π_{CMC} , and $p_{\text{C}_{20}}$) enables knowledge transfer from data-rich to data-scarce targets.

Evaluation on 140 held-out surfactants demonstrates mean $R^2 = 0.848$, with multi-task learning benefiting data-scarce properties the most. Individual R^2 values: 0.902 (p_{CMC}), 0.809 (γ_{CMC}), 0.805 (Γ_{max}), 0.868 (A_{min}), 0.808 (π_{CMC}), and 0.898 ($p_{\text{C}_{20}}$).

Multi-task learning benefits are most pronounced for data-scarce properties: Γ_{max} (+15.0%) and A_{min} (+15.6%) over single-task models of identical architecture, while data-rich properties (p_{CMC} and γ_{CMC}) achieve comparable accuracy. Ablation studies confirm that ensemble averaging contributes +3.3% and that the learned temperature encoder outperforms simple concatenation by 1.9%; temperature confers negligible incremental benefit to traditional ML baselines (≤ 0.001), confirming that the GNN representation and multi-task architecture, not temperature alone, drive performance gains.

The 60-model ensemble provides prediction uncertainty with 75.5% empirical coverage at the nominal 95% CI (mean ECE = 0.134), indicating systematic overconfidence that identifies a clear direction for future calibration improvements. The uncertainty–error correlation ($\rho = 0.58$) remains practically useful for identifying predictions requiring experimental verification. While specialised single-property models achieve marginally higher accuracy for individual targets, SurfMT-GNN uniquely provides comprehensive multi-property prediction with temperature modelling and uncertainty quantification across all six properties, essential for practical multi-objective surfactant optimisation. Sub-second inference enables high-throughput screening exceeding 1000 molecules per minute.

Future work should address post-hoc uncertainty recalibration, surfactant mixtures, physics-informed constraints, expanded training data for underrepresented classes, and generative design integration. SurfMT-GNN establishes a foundation for uncertainty-aware surfactant property prediction, accelerating rational design for sustainable applications.

Author contributions

Al-Futini Abdulhakim Nasser Ali: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing – original draft. Huang Hexin: conceptualisation, supervision, project administration, funding acquisition, writing – review and editing. Ali A. AL-Bakhrani: methodology, software, writing – review and editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The original SurfPro dataset is publicly available from Hödl *et al.*⁴ at <https://github.com/BigChemistry-RobotLab/SurfPro> and archived on Zenodo (DOI: 10.5281/zenodo.14931937). The complete SurfMT-GNN implementation including molecular featurisation scripts, model architecture, training code, trained model weights, and processed datasets, is openly available at <https://github.com/albakhrani/SurfMT-GNN>, with the version of record accompanying this article archived on Zenodo (DOI: 10.5281/zenodo.20761125). All molecular structures and property values used for training and evaluation are provided in machine-readable CSV format in the repository. The code is released under the MIT Licence.

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